Encryption and Decryption Using Caesar and Vigenère Ciphers

Objective:

The objective of this activity is to demonstrate the process of encrypting and decrypting messages using classical cryptographic techniques, specifically the Caesar Cipher and the Vigenère Cipher. The activity also aims to verify the results using online encryption tools to ensure accuracy.

Verification Tools Used:

CAESAR CIPHER: https://www.codbolt.com/tools/caesar-cipher

VIGENÈRE CIPHER: https://www.prodevtool.com/tools/vigenere-cipher

PART 1: CAESAR CIPHERS

1. ***Plaintext***: DATA SECURITY IS IMPORTANT ***Key Shift***: 3

Encryption Process:

PLAINTEXT	CIPHERTEXT	PLAINTEXT	CIPHERTEXT
D	G	I	L
A	D	S	V
T	W	I	L
A	D	M	P
S	V	P	S
E	H	O	R
C	F	R	U
U	X	T	W
R	U	A	D
I	L	N	Q

T	W	T	W
Y	B		

Therefore, the ciphertext is GDWD VHFXULWB LV LPSRUWDQW.

Decryption Process: *(Shifting the ciphertext three positions backward)*

CIPHERTEXT	PLAINTEXT	CIPHERTEXT	PLAINTEXT
G	D	L	I
D	A	V	S
W	T	L	I
D	A	P	M
V	S	S	P
H	E	R	O
F	C	U	R
X	U	W	T
U	R	D	A
L	I	Q	N
W	T	W	T
B	Y		

Therefore, the plaintext is DATA SECURITY IS IMPORTANT.

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2. ***Plaintext***: PROTECT YOUR PASSWORD

Key Shift: 5

Encryption Process:

PLAINTEXT	CIPHERTEXT	PLAINTEXT	CIPHERTEXT
P	U	R	W
R	W	P	U
O	T	A	F
T	Y	S	X
E	J	S	X
C	H	W	B
T	Y	O	T
Y	D	R	W
O	T	D	I
U	Z		

Therefore, the ciphertext is UWTYJHY DTZW UFXXBTWI.

Decryption Process: *(Shifting the ciphertext five positions backward)*

CIPHERTEXT	PLAINTEXT	CIPHERTEXT	PLAINTEXT
U	P	W	R
W	R	U	P
T	O	F	A
Y	T	X	S
J	E	X	S
H	C	B	W
Y	T	T	O
D	Y	W	R
T	O	I	D
Z	U		

Therefore, the plaintext is PROTECT YOUR PASSWORD.

PART 2: Vigenère Ciphers

1. ***Plaintext***: CYBER SECURITY IS ESSENTIAL ***Keyword***: KEY

Key Expansion:

Plaintext: CYBERSECURITYISESSENTIAL

Key : KEYKEYKEYKEYKEYKEYKEYKEY

Encryption Process:

PLAINTEXT	P	KEY	K	P + K (MOD 26)	CIPHERTEXT
C	2	K	10	12	M
Y	24	E	4	2	C
B	1	Y	24	25	Z
E	4	K	10	14	O
R	17	E	4	21	V
S	18	Y	24	16	Q
E	4	K	10	14	O
C	2	E	4	6	G
U	20	Y	24	18	S
R	17	K	10	1	B
I	8	E	4	12	M
T	19	Y	24	17	R
Y	24	K	10	8	I
I	8	E	4	12	M

S	18	Y	24	16	Q
E	4	K	10	14	O
S	18	E	4	22	W
S	18	Y	24	16	Q
E	4	K	10	14	O
N	13	E	4	17	R
T	19	Y	24	17	R
I	8	K	10	18	S
A	0	E	4	4	E
L	11	Y	24	9	J

Therefore, the ciphertext is MCZOVQOGSBMRIMQOWQORRSEJ.

Decryption Process:

CIPHERTEXT	C	KEY	K	C - K (MOD 26)	PLAINTEXT
M	12	K	10	2	C
C	2	E	4	24	Y
Z	25	Y	24	1	B
O	14	K	10	4	E
V	21	E	4	17	R
Q	16	Y	24	18	S
O	14	K	10	4	E
G	6	E	4	2	C
S	18	Y	24	20	U
B	1	K	10	17	R

M	12	E	4	8	I
R	17	Y	24	19	T
I	8	K	10	24	Y
M	12	E	4	8	I
Q	16	Y	24	18	S
O	14	K	10	4	E
W	22	E	4	18	S
Q	16	Y	24	18	S
O	14	K	10	4	E
R	17	E	4	13	N
R	17	Y	24	19	T
S	18	K	10	8	I
E	4	E	4	0	A
J	9	Y	24	11	L

Therefore, the plaintext is CYBERSECURITYISESSENTIAL.

REFLECTION:

This activity gave me a deeper appreciation for how classical cryptographic methods paved the way for modern cybersecurity. By exploring these early encryption techniques and understanding how they can be deciphered, I gained insight into why contemporary cryptographic algorithms must be far more sophisticated and resistant to analytical attacks. The experience highlighted the importance of continuously advancing encryption methods to maintain secure communication in today's digital landscape.